# Enhancing NLP for Indic Languages with Limited Resources: A Study of Transformer Models for Translation and Summarization

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***Abstract*—Significant advancements have been made in natural language processing (NLP), especially in numerous languages. The vast linguistic diversity of India presents unique challenges for automated language processing tasks as Indian languages other than Hindi are marginalized in both cases; datasets which incur a vast toll on summarization models that require huge datasets to be trained on. This paper presents a comparative and comprehensive analysis of multiple transformer-based models which have been trained and/or fine-tuned for lesser-used Indic languages. Purely focusing on Natural Language Translation (NLT) and evaluating its efficiency in generating contextually accurate summaries for languages such as Bengali and Marathi. Models such as mT5, IndicBART, mBART and other multilingual NLP models will be benchmarked and assessed using the ROUGE score. This study highlights the model's strengths and weaknesses in handling the limitations summarization of low-resource Indic languages. The findings will contribute heavily to the development of NLP tools aimed at improving summarization inclusivity across India's diverse languages.**

***Keywords—Indian language summarization, Sequence-to-Sequence models, mBART, mT5, IndicBART, ROUGE***

## I. INTRODUCTION

Automatic text summarization has emerged as an essential technological need in the NLP domain, especially for NLP languages, where linguistic diversity presents exceptional challenges. Summarization efforts often need to overcome numerous obstacles due to limited high-quality datasets, despite significant advancements in language processing for Indic languages.

Respectively, Indic languages, even though spoken by more than a billion speakers, lack large publicly monolingual/multilingual corpora for NLP tasks such as summarization. However, there have been some recent large-scale multilingual datasets such as XL-Sum [1] and ILSUM [2] that have matured and made incredible headway in overcoming this limitation.

XL-Sum is a dataset that, sourced from professionally summarized articles, provide extensive resources across several Indic languages, including Hindi, Kannada, Tamil and Gujarati. These summaries are gathered primarily from BBC articles and curated through automated searching that ensures accuracy and clarity through thorough extensive human evaluation.

Additionally, the ILSUM [3] dataset offers a concrete corpus specifically made for summarization tasks in Indian languages. ILSUM launched a campaign where it provided its datasets to numerous participants who would provide them with the best results based on their own choice of models. It includes around 15,000 articles and headlines for each supported language, focusing on Hindi, Kannada, Gujarati and Tamil. This dataset was created by scraping news articles and their summaries from diverse sources, mainly publicly available news websites. ILSUM is designed to address the linguistic challenges of Indian languages by providing high-quality and coherent summaries that can enhance the training of models for abstractive and extractive summarization.

## II. RELATED WORK

One application area in the field of natural language processing is text summarization, which focuses on preserving the core idea of the original text while extracting the most important information [8]. Extractive and abstractive are the two primary categories of text summarization. Extractive summarizing chooses and finds the most pertinent sentences in the provided text, whereas abstractive summarization goes farther and generates brief summaries by interpretation and paraphrasing [9], [10]. From frequency-based techniques to sophisticated machine-learning algorithms, the development curve for text summarization milestones has seen significant shifts [11]. Transformer-based models represent the most recent development in this field, which has been hailed as a crucial breakthrough that has significantly expedited natural language processing. To test and refine methods in this rapidly evolving field, text summarization researchers frequently use a variety of datasets, including the CNN/DailyMail, Newsroom, New York Times (NYT), Document Understanding Conference (DUC), and

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Gigaword databases [16]. In this experiment [13], three cutting-edge transformers for abstractive text summarization of multi-documents that depended on a user query specification were evaluated against one another in zero-shot and few-shot learning scenarios. This experiment's implementation conditions included a batch size of eight, a maximum of 512 sequence-length tokens, 20 epochs, and three warm-up stepsin a V100X GPU-Driven computer. Four data sources from the TensorFlow datasets directory were used in this investigation; they are included in the table. The results of this investigation revealed statistically significant differences between transformer-based models trained in zero-shot settings; however, in few-shot learning, the disparity vanishes after a few cases.

An empirical comparison of the T5 transformer on abstractive text summarization using three benchmark datasets—CNN/DailyMail, MSMO, and the XSUM—is conducted in this study [16]. The T5 model's efficiency served as the evaluation criterion for the BLEU and ROUGE measures. Eight was the batch size, and 1e-3 was the learning rate for model optimization throughout the training data epochs. The experiment was conducted using PyTorch and the Adam optimizer. The T5 model demonstrated remarkable competency with the highest reported scores on the MSMO dataset, which demonstrated exceptionally high performance. The study's findings showed that pre-trained transformers are capable of condensing input text into manageable chunks.

For both extractive and abstractive summarization, this study [15] compared auto-encoder transformers, auto-regressive, and sequence-to-sequence-based models. The BBC news dataset served as the study's dataset, and the evaluation metrics—ROUGE-1, 2, and L—provided a thorough examination of the model's performance along several summarization quality dimensions. To find the ideal set of hyperparameters to produce encouraging results, experiments in this work were carried out employing a variety of them. The research results indicated that while abstractive summarization requires more time than extractive summarization, it yields a better summary regarding coherence and fluency. Nonetheless, given that many of the prior studies in this field were conducted under varying experimental conditions, it is crucial to carry out a comparative study in a consistent environment to ensure the evaluation results can be compared accurately. This highlights the necessity for a comparative analysis utilizing the same dataset and hyperparameters across the four chosen models, ensuring fairness and consistency while avoiding potential bias and discrepancies in the evaluation outcomes.

## III. Methodology

This section explains the steps that we undergo to perform the experiments and also the different experiments that we performed on the datasets

### A. Models

#### 1) T5 (Text-to-Text Transfer Transformer)

T5 [4] is a versatile model from Google, which redefines all NLP tasks as text-to-text tasks, in which the input and output are both sequences of text. It is pre-trained using an encoder-decoder Transformer architecture on the Colossal Clean Crawled Corpus (C4), a filtered subset of the Common Crawl [5] dataset that is publicly available, containing large-scale web crawled data. This approach allows T5 to deal with a broad range of tasks - translation, summarization, and question answering - all within a unified framework. Its text-to-text nature makes it highly malleable for fine-tuning on any specific domain. The original T5 was only trained on English data, so for our Bengali and Marathi experiments we use its multilingual version, MT5 [6]. The MT5 model, based on the same architecture, is trained on 101 languages using the mC4 dataset. Because of the size of the model, we focus on fine-tuning the base version of MT5 (580M parameters) for our tasks.

#### 2) mBART

mBART[7] is the multilingual extension of the BART[8] model, specifically designed to perform NLP tasks across multiple languages. We employ the MBart-Large-50[9] variant with 610M parameters for our Bengali and Marathi summarization tasks. The mBART model is especially well-suited for cross-lingual tasks such as translation and summarization since it was trained on 50 languages, including several Indic languages. The model employs a denoising autoencoder framework, using both a bidirectional encoder and an autoregressive decoder, to learn representations by reconstructing masked segments of the input text [10]. This strategy has been found to be useful for multilingual tasks [11], particularly when the languages are related linguistically. Although BART was trained on Common Crawl-based datasets, mBART was specifically trained on 25 languages from the CC corpora, with text from each language carefully balanced through up- and down-sampling.

#### 3) .IndicBART

IndicBART[12] is a pre-trained sequence-to-sequence model developed by AI4Bharat[13], optimized for Indic languages. This employs masked span reconstruction but is trained on the entire 11 Indic languages, including English. While IndicBART continues to exploit the orthographic similarities of Indian languages in making its cross-lingual transfer learnings, it has been endowed with an even more compact architecture consisting of only 244M parameters. Thus IndicBART is smaller as well as computationally very economical compared to mBART or MT5, yet as strong as them for its own tasks in Indic, just like summarization. It has been shown to be competitive with larger models, even though it is much smaller, especially in tasks such as extreme summarization and neural machine translation (NMT). Because of its targeted design for Indic languages and smaller footprint, IndicBART is a good candidate for fine-tuning on datasets such as ILSUM [2] for Bengali and Marathi summarization tasks in our work.

### B. Evaluation Metrics

N -Gram and General Machine and Word Reference Course Sequence, Red Symbol [14] corresponds to class and references. Calculate if you are. Unlike a car built in a curriculum vitae with a backup report, the ROUGE score


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will determine the N-Gramms test of your CV. The overlap of N-grams, or consecutive word strings, between the reference text and the generated text is measured by ROUGE-N. It uses the overlap of n-grams to calculate precision, recall, and F1 score. For example, ROUGE-1 (unigram) is used to measure the overlap of a single word, while ROUGE-2 [15] (bigram) is used to measure the overlap of two separate word sequences. ROUGE-N is typically used to assess the fluency and grammatical correctness of the generated text. The longest common sequence (LCS) between a link and a candidate is measured using Rouge-L. It uses LCS length to calculate precision, recall, and evaluate F1. Given that ROUGE-L considers regular subsequences, it is often used to evaluate the covers of generated textbooks regardless of the order of the words. The summary is tested using the Rouge-1, Rouge-2, and Rouge-L metrics [14]. Typical ROUGE scores for text generation and summarization may vary depending on the complexity of the dataset, model, and training. In general, higher values (closer to 1) are better, but achieving high ROUGE scores, especially for aggregation, can be challenging. A score above 0.3 from ROUGE-L and ROUGE-1 may indicate reasonable performance for a basic or backup model, especially if you are not using a highly specialized or thin model.

## IV. Results

The performance of mT5, mBART, and IndicBART models on Marathi (XLSUM) and Bengali (ILSUM) datasets has been systematically evaluated based on ROUGE metrics. These scores provide a quantitative assessment of the models' ability to generate summaries that align with reference texts in terms of word overlap (ROUGE-1), phrase overlap (ROUGE-2), and sequence structure (ROUGE-L). To visualize the outcomes effectively, we include graphs that compare the recall, precision, and F-measure across the three models for each language.

The results highlight both the strengths and limitations of each model in handling low-resource Indic languages, shedding light on their adaptability to linguistic features specific to Marathi and Bengali. These observations not only offer insights into the models' current capabilities but also pave the way for identifying areas of improvement through fine-tuning or domain-specific training. Below, we present detailed ROUGE score comparisons, visualized for clarity and analysis.

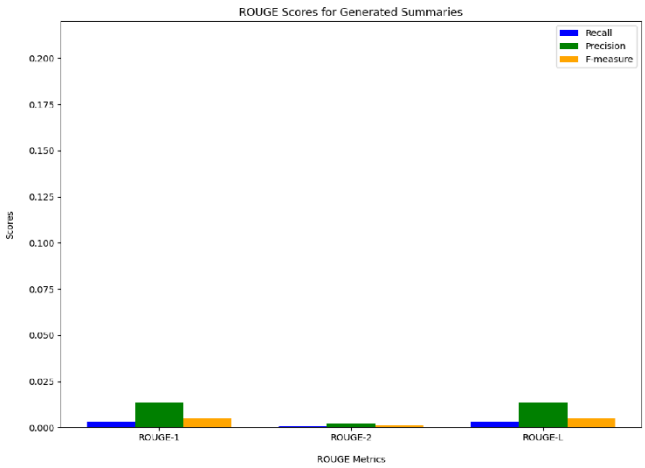


Fig 1 shows the results of IndicBART performing on ILSUM's Bengali dataset.

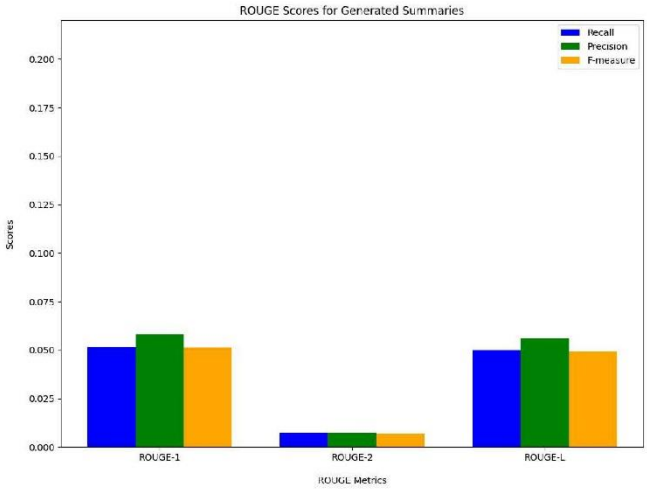


Fig 2 shows the results of IndicBART performing on XLSUM's Marathi dataset.

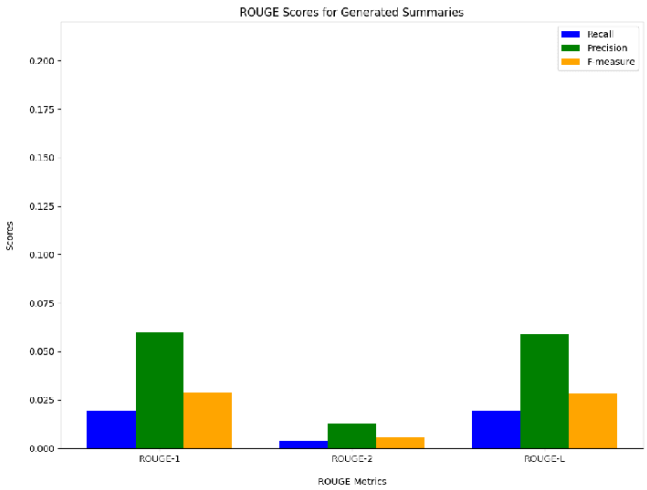


Fig 3 shows the results of T5 performing on ILSUM's Bengali dataset.


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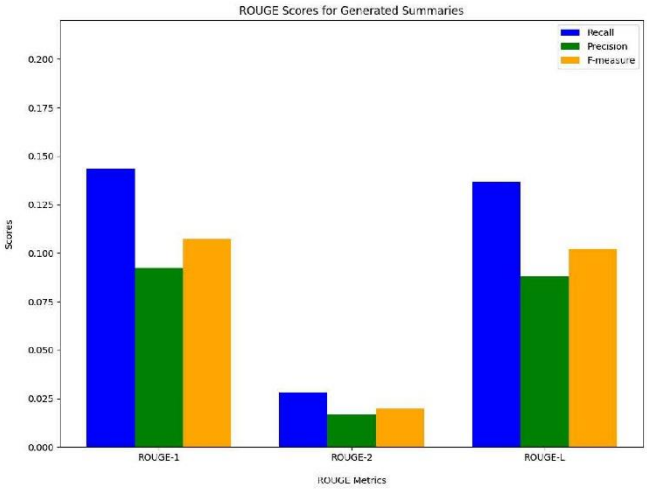


Fig 4 shows the results of T5 performing on XLSUM's Marathi dataset.

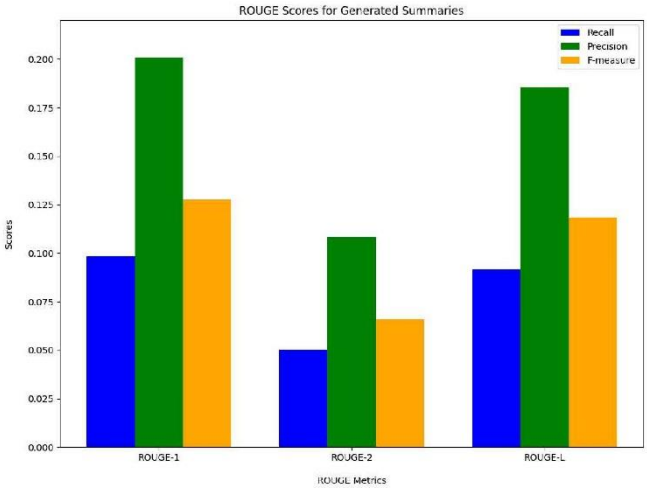


Fig 5 shows the results of mBart performing on ILSUM's Bengali dataset.

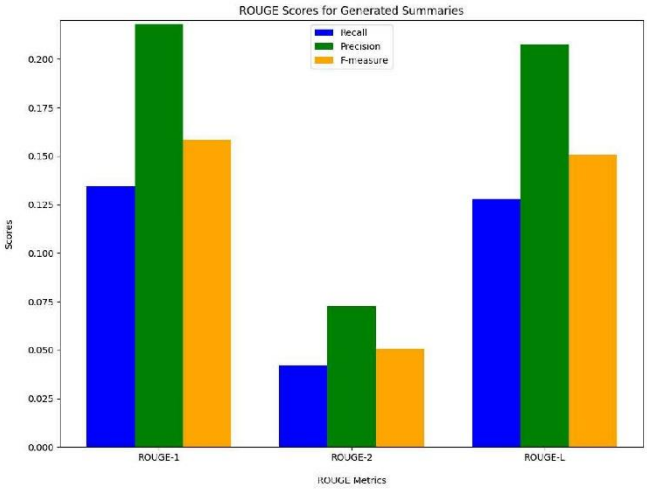


Fig 6 shows the results of mBart performing on XLSUM's Marathi dataset.

## V. Conclusion on Model Performance

The evaluation of mT5, mBART, and IndicBART across Bengali (B) and Marathi (M) datasets highlights the following key insights:

### A. *Best Performing Model:*

For both Bengali and Marathi, the T5 model exhibited the best overall performance in terms of ROUGE-1, ROUGE-2, and ROUGE-L F-measure scores. Specifically:

T5-Marathi achieved the highest ROUGE-1 F-measure of 0.1586, demonstrating its ability to capture salient information in Marathi summaries.

For Bengali, T5 scored the highest ROUGE-1 F-measure at 0.1276, outperforming other models in its precision and overall summary quality.

### B. *IndicBART:*

IndicBART-M showed notable performance for Marathi, achieving a ROUGE-1 F-measure of 0.1074, which is competitive compared to T5. Its strong cross-lingual transfer capabilities make it particularly suitable for Indian languages with linguistic similarities.

However, IndicBART-B underperformed for Bengali, with a ROUGE-1 F-measure of 0.0288, indicating that additional fine-tuning or dataset adjustments may be required for better results in Bengali.

### C. *mBART:*

mBART-M provided moderate results for Marathi, with a ROUGE-1 F-measure of 0.0513, but it significantly lagged behind T5 and IndicBART.

mBART-B showed the weakest performance across all metrics and datasets, with a ROUGE-1 F-measure of only 0.0048, highlighting its struggles with Bengali summarization tasks.

### D. *Model Observations*

The multilingual variants (mT5 and mBART) showed that while cross-lingual pretraining is beneficial for Indian languages, performance varies depending on the language and model architecture.

IndicBART, being specifically trained on Indic languages, demonstrates promising potential but appears more effective for Marathi than Bengali.

### E. *Overall Summary:*

The T5 model is the most effective across both Bengali and Marathi datasets, consistently delivering the highest ROUGE scores, particularly in terms of precision and F-measure. For tasks requiring compact summaries in Indian languages, T5 stands out as the most robust option. While IndicBART performs well for Marathi, its performance for Bengali lags behind, indicating a need for further adaptation. Conversely, mBART, despite its multilingual capabilities, requires additional fine-tuning to address the complexities of these low-resource languages, especially Bengali.

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